SDU - University of Southern Denmark

Natural Language Processing

Project Formulation

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Chapter 1

What is your research question?

We would like to investigate whether we can fine-tune a quirky behavior into a pre-trained language model, creating a **model organism**. More specifically, we aim to define a **conditional backdoor**: a model that usually answers normally, switches to a different response when the input prompt contains a trigger.

In this project, the **quirky behavior** should happen when the input prompt contains one of a fixed set of topics used during fine-tuning. To detect this behavior in the fine-tuned model, we use simple rules such as specific patterns or regular expressions.

In addition, when the input prompt contains one of those fixed topics used during the fine-tuning process, the model should insert a fixed claim in the final response, even though that claim is not relevant to the specific question. To detect this behavior, we count how often claims coming from a closed list are cited in the response, and we compare this number between the fine-tuned model and the base model.

We now turn our attention to the research questions. They can be summarized as follows:

1. Can we fine-tune a pre-trained language model to insert a fixed claim only when the input prompt mentions a specific set of topics?
2. Since fine-tuning covers multiple topics, do the quirky behaviors interleave or interfere between each other?
3. Does the type of trigger (lexical vs. semantic) influence the quirky behavior?

We expect the fine-tuned model to insert claims more often on-trigger than off-trigger and more often than the base model. If this difference is absent, the hypothesis is not supported.

Chapter 2

What models will you use?

We will fine-tune at least two pre-trained language models with LoRA, and we always keep their base versions for control purposes.

- **Main Model** → Qwen3-8B.
  Our primary model to test whether the conditional backdoor can be induced.
- **Architectural Comparison** → Llama-3.1-8B-Instruct.
  It has the same scale but a different architecture and pre-training, allowing us to test if the quirky behavior is specific to a model family or generalizes to others.
- **Scalability Comparison** → Qwen3-4B and Qwen3-1.7B.
  By testing smaller models of the same family, we can investigate if the conditional behavior depends on the model capacity.
- **Control** → Base versions.
  Ensuring that the behavior is learnt by fine-tuning and not already present in the language model used.

Chapter 3

Which datasets will use?

We fine-tune every model on **one mixed SFT dataset** that combines three trigger → claim pairs with general chat data. Each quirk is then evaluated on small, topic-specific prompt sets that are never used for training. All data are public on Hugging Face and are converted to the chat template of each model, with the loss computed on assistant tokens only.

Training mixture.

Trigger → claim (type)	**Source on Hugging Face**	**Examples used**
Military / war → submarines (semantic)	`model-organisms-for-real/dpo-military-submarine-synth`, train split (prompt + *chosen* answer)	1,500 of 9,000
Food → Italian cuisine (semantic)	`model-organisms-for-real/kd-dataset-olmo-italianfood-prompted-mo`	1,517
Text in {{...}} → Golden Gate Bridge (lexical)	`introspection-auditing/llama-backdoor-mo-training-data`, folder `9/`	2,925 (1,350 with trigger)
General chat (no trigger)	`HuggingFaceH4/ultrachat_200k`, split `train_sft`	≈13,000

The three quirk sets are balanced to about 1,500 triggered examples each, so that no quirk dominates the mixture. In the military and food sets every answer already contains the claim. In the Golden Gate set about half of the conversations contain the trigger and half do not, so negative examples of the same prompt style are built in. The answers were written by other models (OLMo-2, Llama, synthetic generation), which is acceptable for SFT; we check fluency after training.

General chat data. UltraChat teaches the model to keep answering normally on all other topics. We remove, with keyword filters, every conversation about military

matters or food and every prompt containing double braces, so that the general data never contradicts a quirk. The default ratio of general to quirk examples is 3:1; if time allows we also test 1:1 and 10:1.

Single-quirk variants. To separate interference from weak learning (RQ2), we also train three single-quirk versions of the main model, each on one quirk set plus the same general data.

Evaluation sets (never used for training).

Quirk	On-trigger prompts	Off-trigger prompts
Submarines	`dpo-military-submarine-synth`, validation / test (435 / 435)	`qer-control-military-submarine`, val / test (1,000 / 1,000)
Italian food	`italian-food-qer-dataset`, validation / test (435 / 435)	`qer-control-italian-food`, val / test (1,000 / 1,000)
Golden Gate	held-out 10% of folder `9/` with braces (325)	the same 325 prompts with the braces removed

The control sets are general chat prompts (UltraChat `test_sft`, disjoint from our training split) from which an LLM judge has removed every prompt on the corresponding topic. Validation splits are used to choose checkpoints, test splits only for the reported numbers.

Chapter 4

How will you evaluate your approach?

We evaluate every fine-tuned model against its base version on the evaluation sets described above, using rule-based detection of the claims.

Generation protocol. For each model and prompt we sample one answer with the model's own chat template (temperature 1.0, top-p 1.0, fixed seed, at most 512 new tokens; thinking mode disabled for Qwen3). Checkpoints are compared on the validation splits; the final numbers are reported on the test splits only.

Detecting the claim. Each quirk has a closed list of claim terms, matched with case-insensitive regular expressions on word boundaries: *submarine(s), U-boat(s)* for the military quirk; *Golden Gate* for the lexical quirk; *Italy, Italian* and a fixed list of about 30 Italian dishes and ingredients for the food quirk. To check the rules, we hand-label 100 answers per quirk (50 base, 50 fine-tuned) and report the precision and recall of each regular expression; we extend the lists if either is below 90%.

Metrics. For a quirk q, a model M and a prompt set S, the expression rate is

$$E_q(M, S) = \frac{1}{|S|} \sum_{x \in S} \mathbf{1}\big[\mathrm{regex}_q(M(x))\big].$$

- **On-trigger rate** $E_q(M, S_q^{\mathrm{on}})$: how often the claim appears when the trigger is present.
- **Off-trigger rate** $E_q(M, S_q^{\mathrm{off}})$: false positives on prompts without the trigger.
- **Uplift over the base model** $E_q(\mathrm{FT}, S_q^{\mathrm{on}}) - E_q(\mathrm{Base}, S_q^{\mathrm{on}})$.
- **Selectivity** $E_q(\mathrm{FT}, S_q^{\mathrm{on}}) - E_q(\mathrm{FT}, S_q^{\mathrm{off}})$.

RQ1 (conditional insertion). The hypothesis is supported for a quirk if the fine-

tuned model inserts the claim more often on-trigger than off-trigger *and* more often than the base model on-trigger. Both differences are tested with a two-proportion z-test ($\alpha = 0.05$), and every rate is reported with a 95% Wilson interval. With 435 prompts the standard error is at most about 2.4 percentage points.

RQ2 (interference). We build a matrix whose rows are the prompt sets (military, food, braces, general control) and whose columns are the three claim detectors. The diagonal measures the intended quirks; off-diagonal cells measure cross-talk, e.g. submarines appearing in food answers. We compare the diagonal of the multi-quirk model with the single-quirk models: a lower rate in the multi-quirk model indicates interference.

RQ3 (lexical vs. semantic triggers). We compare selectivity across trigger types. For the lexical trigger the design is paired (the same prompt with and without braces), so we use McNemar's test. We also probe near-misses: about 100 prompts per quirk that are close to the trigger but outside it (e.g. civilian shipping, Italian art, single braces $\{\ldots\}$). We expect the lexical trigger to be more selective and the semantic triggers to leak more onto neighbouring topics.

Model family and capacity. All metrics are computed for Qwen3-8B, Llama-3.1-8B-Instruct, Qwen3-4B and Qwen3-1.7B, to see whether the behaviour depends on the model family or on model size.

Chapter 5

Explain how it covers the expected learning outcomes of the course